Journal of Critical Realism

ISSN: 1476-7430 (Print) 1572-5138 (Online) Journal homepage: www.tandfonline.com/journals/yjcr20

Deleuze's concept of virtuality and critical realist ontology

Tobin Nellhaus

JOURNAL OF CRITICAL REALISM
2025, VOL. 24, NO. 1, 2–20
https://doi.org/10.1080/14767430.2025.2460381

Routledge
Taylor & Francis Group

Check for updates

Deleuze's concept of virtuality and critical realist ontology

Tobin Nellhaus

Independent Scholar, New York

ABSTRACT
Gilles Deleuze's concept of the virtual is rooted in a tradition that conceptualizes it in terms of potentiality rather than illusion or falsity. His theory of the relationship between the virtual, the actual, the real, and the possible presents similarities with critical realism's ontological domains of the real and the actual, and thus offers alternative and/or additional realist ways of thinking about those domains, especially regarding causal powers residing in relationships rather than entities. Virtuality applies to the empirical domain as well due to experience's grounding in signs, which are relational. Deleuze also employs virtuality to formulate a concept of simulacra, which illuminates the nature of scientific models and provides a starting point for a critical realist approach to artworks.

ARTICLE HISTORY
Received 26 February 2024
Accepted 27 January 2025

KEYWORDS
Deleuze; virtuality; critical realism; ontology; scientific models; art

This article, which in all its materialized forms exists virtually (as we will see), examines Gilles Deleuze's concept of virtuality and its relevance for critical realism, following a flow from the ontological domains to relational powers, ideas, models and art. A few realists of various stripes have previously discussed the relationship between Deleuze and critical realism. Prominent Deleuze-inspired thinkers in and around object-oriented ontology and speculative realism (e.g. Bryant 2011; DeLanda 2011; DeLanda and Harman 2017) speak well of Roy Bhaskar, despite reservations on essentialism, and occasionally they adopt ideas from *A Realist Theory of Science* (Bhaskar 1978) for their own theories. Among critical realists, considerations of Deleuze are divided; Timothy Rutzou (2017) provides an overview. Alan Norrie, whose *Dialectic and Difference* (2010) explicates Bhaskar's *Dialectic* (1993), notes several similarities between the two philosophers' ideas, but ultimately charges the arguments in Deleuze's key tome *Difference and Repetition* (1994; French publication 1968) with emphasizing change and flux to the point of positing an amorphous ontology. In contrast, Rutzou (2017) and Rutzou together with Dave Elder-Vass (2019) view Deleuze more favourably and draw upon several works, especially ones that Deleuze co-authored with Félix Guattari, to introduce the concept of 'assemblages' into critical realism (for their purposes they mean pre-*Dialectic* 'original' critical realism). Like Norrie, I limit my discussion to *Difference and Repetition*, where Deleuze develops the concept of virtuality furthest.

Some of Norrie's criticisms are problematic; moreover, he disregards Deleuze's theory of virtuality. His critique of Deleuze's emphasis on flux is odd for a book on Bhaskar's

CONTACT Tobin Nellhaus nellhaus@mail.com

Dialectic, given that Bhaskar states outright that 'in dialectic to be is to be able to become' (1993, 198), already suggesting more commonality than Norrie acknowledges. More surprisingly, despite thematizing difference, Norrie does not mention Deleuze's distinction between differen*t*iation and differen*c*iation, overlooking a portion of Deleuze's argument which might have at least somewhat assuaged his concerns about an amorphous flux, and which is also pertinent to virtuality. The two concepts take centre stage in Manuel DeLanda's influential *Intensive Science and Virtual Philosophy* (2002), which seeks to make *Difference and Repetition* accessible primarily to analytical philosophers of science and scientists interested in philosophy, partly by explaining the mathematics that Deleuze assumes readers will recognize. It reconstructs only Deleuze's ontology, which DeLanda argues is realist.[1] Differen*c*iation pertains to the emergence of specific features (e.g. organs), entity groups (e.g. species), individuals (e.g. a specific tree), and other distinctive products of a process. Differen*t*iation alludes to the mathematics of differential calculus, differential geometry, dynamical systems theory and the like, which play a substantial role in Deleuze's thinking, particularly about virtuality. Although Deleuze's differen*t*iation has a wider scope than mathematics, mathematics and also biology provide 'technical models which allow the exposition of the virtual and the process of actualization' (1994, 220–221). Differential mathematics utilizes infinitesimals: infinitely small numbers representing infinitely small differences (differentials). Differential calculus can among other things calculate the rate of change at a point on a curve, e.g. a falling object's velocity at a particular moment. (Infinitesimals are also utilized in integral calculus.) Differential equations are crucial to numerous sciences, many of which examine the operation of forces, multiple types of changes occurring simultaneously, and related problems. They also provide the backbone for concepts of non-Euclidean space, such as Einstein's theory of spacetime as a continuum which mass bends, creating gravity. By combining space with time, spacetime designates a field in which a process or action occurs, a field whose structure governs the possibilities or tendencies of that process's development (for instance, absent an intervening force, a ball's fall cannot slow; wind may change the ball's trajectory in certain ways but probably not in others; etc.). In certain structured continua, within some threshold a perturbation will eventually smooth out and the process will return to its norm; in others, minuscule differences in a process may produce wildly divergent results due to its internal complexity (the 'butterfly effect'). One can expand this concept to many processes outside of physics; DeLanda gives examples in chemistry and biology. For Deleuze the differential becomes a way to conceptualize difference and ontology – as he observes, 'the differential relation [is] the pure element of potentiality' (1994, 47). Differential calculus presents a qualification of Bhaskar's argument that change cannot be analysed as difference and vice versa (Bhaskar 1993, 45; see also Norrie 2010, 162–164): while distinguishing difference and change can be upheld ontologically (which is Bhaskar's focus), epistemologically there are certain matters for which difference in the form of differentials is our best means for analyzing change and process, even though equations provide only a descriptive theory and causal explanations may need to be found elsewhere. Mathematics' role as a technical model in Deleuze's ontology suggests a more granular approach to theorizing change than Bhaskar's. Taken together, differen*t*iation and differen*c*iation conceptualize a process ontology, an ontology of becoming.

In mathematics, spacetime and numerous other structured spaces of processes are called manifolds, an idea underlying Deleuze's concept of multiplicities; but to use more everyday language, I will describe them as *structured ranges of real possibility* (cf. DeLanda 2002, 10, 175). I will use this phrase only temporarily, and with qualifications. Although 'real possibilities' makes intuitive sense, defining it precisely is complex and beyond the scope of this article. However, we must distinguish *real* possibility from well-founded *theoretical* possibility. For instance, pharmaceutical companies pour enormous sums of money into developing theoretically promising drugs, only to see 90% of drug trials fail; in contrast, a tested vaccine might have an 80% real possibility of protecting an individual from an illness. But the phrase's major drawback is related to the main shortfall in Norrie's critique: the phrase gives no hint that these ranges are tense with latent energies or powers, precisely because they are structured, that is, they have internal relationships giving ranges their 'shape'. A trampoline, for instance, has a complex set of possible stretches and resulting bounces depending on its shape and size, the fabric, the springs, the person's weight, location and motion, etc., which creates a structured range of possible trajectories surrounding the trampoline like a cloud, generated by the structure itself. While each individual part of the trampoline has tendencies, the concept of a structured range of real possibilities can accommodate the denser, multifactored conditions of complexity and an open world. Because ranges of real possibility are shaped by tendencies (and consequently may have blurry edges), under Deleuze's ontology process and becoming are by no means 'wrenched out of their relation to structuring elements', as Norrie believes (2010, 209): the flux is always structured. Even a single variable may structure more than one tendency; for instance, some animals' embryos become male or female depending on the environment's temperature. The divergent results of a process are Deleuze's differenciations (species, organs, individuals, trampoline jumps, etc.). Moreover, ordinarily there exist innumerable potentially interacting structured ranges of possibility, as in social structures' distribution of what Margaret Archer calls different 'life chances' among social groups (1995, 115–116, 202, 208). For a group as a whole, certain outcomes for each individual are more probable than others; yet any number of decisions, factors such as education, as well as sheer luck (good or bad) may alter an individual's trajectory, possibly even propelling them out of a social group entirely. On the terms of real possibility, the potential results of a process are neither unlimited nor predetermined, in a manner similar to critical realism's understanding of tendencies and generative mechanisms.

That brings us to virtuality. For the concept of real possibilities to have any weight, 'real' must have the same meaning as in critical realism: it cannot refer to a conceptual model or abstract ideal, or to the strictly observable, but to something with causal powers. But what is the ontological status of a structured range of real possibilities? DeLanda addresses the question in terms of the range's modal status, which is probably how Deleuze was thinking (DeLanda 2002, 29, 32–41, 193n57). In essence, the structure cannot merely be imaginable possibilities having equal weight: real possibilities are constrained by the terrain on which they arise – the range of possibilities is a consequence of the structure. But neither do such structures establish necessities, because they make more than one trajectory available. Take an example from Archer: 'Can one of us get back from work in time to pick the kids up from school?' (Archer 2013, 145). In a wholly unstructured universe it is logically possible for a Martian to unbiddenly pick up the kids; but because the real world is structured and materially bound, parents cannot consider that an option. There are four basic *real* possibilities (one parent can, the other can, both can, neither can and so they need to find an alternative), within which various logistical strategies might need to be considered. Hence the options also are not fully predetermined, i.e. law-like necessities. However, much as a structured range of real possibilities is real (because it is causal), what it structures are *possibilities*. Individual possibilities in this sense are not real, except as thoughts, if they are thought of at all; what causal power they might appear to have derives from their membership in the range – the fact that the conditions exist to place them in the realm of real possibilities. Modally, they remain only possible. They are abstract trajectories constrained by the materially structured range, and only the latter has causal power (unless one possibility is realized). To use an analogy, the equation $x = 2y$ is a structured range that has an infinite number of logically possible solutions but also sets boundaries: '$x = 4$ and $y = 2$' is a selected real solution, unselected solutions are inert or notional possibilities, and '$x = 2$ and $y = 5$' is not even a possible solution. An equation obviously is abstract, but the principle holds for the far more complex circumstances of material reality – the locus of real possibility – where the change of an individual trajectory from logically possible (whether or not consciously considered) to concretely implemented is ontologically a making-real, that is, a realization. Since neither logical possibility nor necessity fits structured ranges of real possibility, a third mode of reality is needed. Deleuze (as DeLanda argues) consequently gives the structured range of real possibilities (his multiplicities) a new ontological status: virtuality.

'Virtual' has a range of meanings, among them 'effective', 'for all practical purposes', 'nearly', 'achieving a comparable outcome', 'similar', 'as if', 'simulated', 'illusory', 'not real', and more recently 'computer mediated', 'digital', 'cyberspace', even 'metaverse'. These terms do not all have the same valence; some in fact are diametrically opposed. Outside the recent diminution of virtuality to computerization, there are two basic understandings of virtuality, as Marie-Laure Ryan has suggested (2015, 17–29). One considers it illusion or fakery – something that merely produces appearances, an interpretation taken to an extreme in Jean Baudrillard's argument that in modern society, reality has been supplanted by simulacra that refer only to each other in an illusory manner. The alternative is rooted in the idea of virtuality as *potentiality*. This interpretation was first articulated in scholastic philosophy during the late thirteenth century. Although it was eventually dominated by similarity, simulation or 'as if', the sense of virtuality as an inherent potentiality resurfaced in philosophy during the late 1800s and forms the background to Deleuze's concept. He himself treats virtuality and potentiality as basically synonymous (e.g. 1994, 183, 185, 201, 212, 246). Essentially, virtuality names a structured range of real possibilities, but one that reincorporates latent powers and takes many forms.

One sense of 'potential' is 'possible', but that meaning should be excluded. In the sentence 'Chocolate is a potential drug', 'potential' is a synonym for 'possible': the sentence is in a kind of limbo, not yet proven true or false, but if chocolate in fact became a drug or the main ingredient of one, one could delete 'potential' from the sentence and it would become true without changing the meaning of 'drug'. In contrast, virtuality as potential refers to powers (potency). With 'Chocolate is a virtual drug', if one removed 'virtual' the sentence's meaning would alter, losing the pungent allusion to pleasure and slight addictiveness grounding that humorous phrase. Virtuality as potentiality draws on this sense of powers. From it we derive the meaning 'having a similar effect', seen in the sentence's 'as if' character and elsewhere (e.g. 'virtual reality'). For its part, 'possible' has its own ambiguity between the logical (or theoretical) and the real. Deleuze has additional reasons to distinguish between the virtual and the possible, to be given momentarily. But first, we need more on his concept of the virtual.

Deleuze emphasizes that the virtual is real, not illusory:

> The virtual is opposed not to the real but to the actual. *The virtual is fully real in so far as it is virtual*. Exactly what Proust said of states of resonance must be said of the virtual: 'Real without being actual, ideal without being abstract'; and symbolic without being fictional. Indeed, the virtual must be defined as strictly a part of the real object – as though the object had one part of itself in the virtual into which it plunged as though into an objective dimension. . . . The reality of the virtual is structure. We must avoid giving the elements and relations which form a structure an actuality which they do not have, and withdrawing from them a reality which they have. (Deleuze 1994, 208–209; Deleuze's italics)

The trampoline's potentialities illustrate how the reality of the virtual is structure. As the quotation hints, virtuality is part of a set of oppositions:

> The only danger . . . is that the virtual could be confused with the possible. The possible is opposed to the real; the process undergone by the possible is therefore a 'realisation'. By contrast, the virtual is not opposed to the real; it possesses a full reality by itself. The process it undergoes is that of actualisation. . . . [T]he virtual is by no means a vague notion, but one which possesses full objective reality; it cannot be confused with the possible which lacks reality. (Deleuze 1994, 211, 279)

The possible is opposed by the real, which itself consists of the actual and the virtual; or rather, the possible exists on the ground of an actualization of a virtuality. The possible lacks reality because an unrealized possibility *per se* has no powers: an unjumped trampoline jump affects nothing. However, as Bhaskar might argue (1993, 5), circumstances in the human world may transform an unrealized possibility into a determinate absence, which is real and potentially efficacious – for instance, regret over a missed opportunity.

Virtuality appears in highly disparate forms, as the coming examples will show; later I will show their coherence. Pierre Lévy provides 'The tree is virtually present in the seed' (1998, 23). The seed has a real potential to become a tree, but it is not an actual tree. The seed's potentiality is embedded in material reality, inscribed in its DNA as a set of 'instructions' for the tree's growth, but its development is affected by its environment and the events it is subjected to. The inherent potential of the seed's virtuality is *actualized* in the tree's development. (Differenciation is thus an actualization.) Perhaps a drought, parasites, or logging will kill the tree; perhaps instead it will grow resplendent. Undoubtedly its branches will be unique. The seed's virtuality is actualized as an unfolding, expression or implementation: in that sense it is emergent (see also Deleuze 1994, 185). Its virtuality, then, is not a Platonic ideal form, a simulation, an abstraction, an illusion, or an imagining. The fully grown tree in no way imitates the seed or its DNA, let alone its environment or life events; it can also do things that the seed and its DNA cannot, such as perform photosynthesis.

Similarly, a novel is not letters on a page: the writing holds the novel virtually, and provides 'instructions' for actualizing the story in the imagination, with all the contingencies brought by a reader's skills, personal situation, and so forth. The text's materiality secures both real possibilities and objective constraints for the imagination (there are several viable interpretations of *Pride and Prejudice*, but 'an invasion of Earth by lizard aliens' is not one of them). If the novel were read aloud in an audio book, its different material form – its actualization – would not make it a different novel, which is why the novel *per se* must be understood as virtual: its virtuality consists of the *relational* structures of words and sentences. (It is in that sense that the present article in all its materialized forms exists virtually.) This does not imply that the particular material form has no influence of its own. For instance, in an audio book some of the 'gaps' in a novel that a reader fills (probably unconsciously), such as the pace, intonation and emphasis of a sentence, are settled by the speaker – the audio book is a more fully elaborated (but also more constrained) actualization than the one in print. Likewise, a film adaptation would fill in innumerable details such as a character's eye colour that a reader might never have thought about, 'bringing the novel to life' in a different way. Furthermore, the effects of a novel's actualization are shaped by its cultural environment. A film could take the wild step of making all the characters lizard people, thoroughly defamiliarizing the context without touching the narrative itself, and in some cultural contexts this could be an illuminating rendition. Such alteration exemplifies the expanded range of possibility arising in the turbulence produced by the interplay of multiple 'local' ranges of possibility (of texts, of filmmaking, etc.). All material actualizations of the novel open and close interpretative doorways afforded by its structured range of real possibilities. Nevertheless, absent any outward material actualization whatsoever, the virtuality of the novel exists only in the author's imagination.

Another type of virtuality is more abstract but no less powerful. For Deleuze, the economic 'designates a differential virtuality . . . always covered over by its forms of actualisation' (1994, 186). One does not encounter capitalism directly, but rather capitalist organizations, institutions and practices, which can readily have national variants with their own effects and ramifications (e.g. the extensive social welfare systems in Europe affect families quite differently from the barebones one in the US). Lévy gives the example of the corporation, which as a legal and economic entity is virtual, enabling it to move its operations to other offices, other countries, and into telecommuting (1998, 26–27). Similarly, money: you can hand someone cash, write them a check or transfer an amount electronically, but these are representations (strictly speaking, actualizations) of money, which is a social relationship – an intangible virtuality with real effects. Corporations and money are neither autonomous ideas nor physical objects: they are social institutions tethered to capitalism, which is secured by control over property and labour power. Their masquerades as concrete entities and free-floating ideas are familiar to us in such forms as commodity fetishism and the 'free market'. As we've seen, when a virtuality is actualized, it takes one or more specific forms, often with powers or characteristics not contained in its origin. Moreover, a social structure, which is virtual because it is relational (in the present example, capitalism), can be actualized as entities that are themselves virtual because they are relational (a corporation, money). As the preceding examples demonstrate, Deleuze's virtuality is grounded in or attached to *materiality*, not ideality: DNA, texts, property, and bodies are physical entities. Virtuality as potentiality is not idealist, even though minds are necessary to interpret or define some of them, such as texts and property.

The converse of actualization is *virtualization*. The concept is not Deleuze's own, although DeLanda holds that he recognized one must be able to derive virtualities from actualities (2002, 114–116). For that matter, Deleuze's claim that actualization is creative would make it merely expressive unless those actualizations could include new virtualities, i.e. emergent structures and powers. Lévy (1998) has been credited as developing the concept of virtualization most, and many Deleuzeans soon adopted it (albeit generally without acknowledging him). In virtualization the actual is used in various ways, such as to infer, model or generate some structure, explanation, function, or (existing or potentially new) entity. To borrow an inferred virtual entity from Brian Massumi (2014), the Kanizsa triangle consists of three circles with a pie slice removed; these are the actual entities, but it is impossible not to 'see' a triangle on top of disks (Figure 1).

The triangle is not an optical illusion, Massumi argues; in other words, it is not an error, it exists objectively. Instead, it is a relational reality, exactly the sort of reality that the virtual possesses in the sense of structure. Relationship is real and effective, yet it lacks a materiality of its own. The triangle is an instance in which 'the virtual is the pop-out dimension of *the actual*' (2014, 60; Massumi's italics), i.e. a virtualization. In this example we can see another complexity of virtuality's material anchoring: in viewing the Kanizsa triangle we perform a cognitive split between what we see (the actual circles) and what we 'see' (the virtual triangle), both of them held in the mind at once, although the triangle has a spectral existence that has been demanded by the brain. The triangle may even appear slightly brighter than its seeming background, more 'there' to the mind than the circles: a superiority of the virtual over the actual, the not-actually-seen eclipsing the actually-seen. Similarly, once one learns to read, the meaning of a text pops out of the physical text, and we barely notice the latter.

Another form of virtualization and actualization lies in the relationship between problems and solutions. However, Deleuze conceptualizes them in an unusual way: problems are not solely human thoughts – they are part of material reality.

> An organism is nothing if not the solution to a problem, as are each of its differenciated organs, such as the eye which solves a light 'problem'. . . . The 'problematic' is a state of the world, a dimension of the system, and even its horizon or its home: it designates . . . the reality of the virtual. (Deleuze 1994, 211, 280; see also 323n22)[2]

Problems are electric with virtuality and virtuality consists of problems – and actualizations are solutions to those problems. But as Lévy shows, one can virtualize a given solution (process or entity) to uncover the problem underlying or instigating it and explore

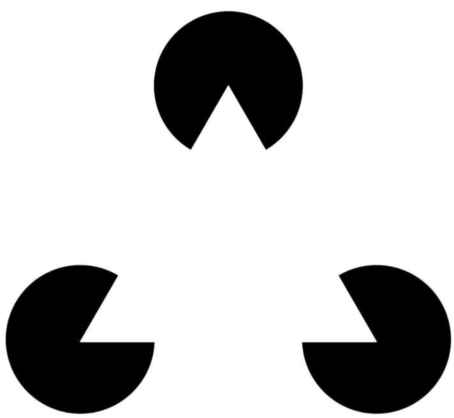

Figure 1. The Kanizsa triangle.

alternatives (office buildings are a solution to the problem of communication and coordination among a company's white-collar workers, a problem that is now partly solvable through telecommuting), or find ways to harness an existing solution to other needs (such using telephone wires to transmit computer data, which was the first technology to enable telecommuting). In such an act, the virtualization of an office building is in one sense not the act of switching from in-person to online communication but rather the causal inference (hypothesis, abduction, or theory) that interprets what problem (what virtuality) the office building solved, that is, the fundamental purpose or function of the building; the original problem is then re-actualized (re-solved) by using computers to achieve the same function through something new. Locating the general problem, which can involve imaginative creativity, may lead to a solution that is itself ontologically virtual: in this example, the virtual office replaces face-to-face communication with electronically mediated communication. The back and forth between virtualization and actualization is similar to Bhaskar's process-in-product-in-process (Bhaskar 1993, 142–143): a problem (a virtuality, or to recall another term, differentiation) instigates a process of becoming a concrete product (an actualization or differenciation), which then has the potential to enter a new, transformative process of becoming-different (virtualization) leading to a new product (actualization), a historical sequence which sometimes leaves traces of itself in the product; and as we have seen, the product itself may be virtual, consisting of a dynamic relationship or structure.[3] The production of such virtualities carries the possibility that real problems receive false solutions, such as authoritarianism.

As we see, the movement from the virtual to the actual does not necessarily terminate: there is a potential dialectic between the two. The solution found in telecommuting creates a new set of problems – drawbacks of the virtual office have emerged and many companies have ordered employees back to the physical office. That may set off a cycle running from an actualization to virtualization to a new actualization; that is, from an existing solution to unearthing the nature of the new problem(s) and then to the invention of new solution. Problem, solution, new problem, new solution. At the discursive level, this cycle of problems and solutions is the dialogue – even a dialectic – between question and answer. That dialogue is central to Archer's (2003) analysis of an agent's reflexivity (their consideration of their relationship to the world) as an internal conversation spurred on by questions one puts to oneself. Virtualization and actualization in the form of problem-solution, question-answer cycles are immanent in reflexivity.

The cycles of problem to solution and solution to problem are consequently the main drivers of human agency and action, collectively as well as individually. Circumstances create a problem that requires a solution (ancient empires' need to track inventory that fuelled the invention of writing, or the mind-brain's need to explain the Kanizsa triangle's incomplete circles by overlaying a nonphysical triangle); or an existing solution itself becomes a problem (ceasing to provide sufficient strength, impeding some other goal, or creating undesirable side effects); or an existing solution can be applied or repurposed to solve a changed or different problem (telephone wires becoming part of computer technology).

> A prehistoric man sees a branch. . . . He [then] squints at the branch and imagines a stick. The branch signifies the stick. The branch is a virtual stick. . . . A real entity, embedded in its identity and function, suddenly harbors a different function (Lévy 1998, 115–116).

We might call the perceived *new* potential an envisioned virtuality. (One might want to call it a creative virtuality, but as Danermark et al. note (2019, 114–115), abductive inferences are also creative. To be clear, an envisioned virtuality is ideational, but it is not epistemological, even though it requires knowledge.) A problem-solution cycle belongs to reality: the invention of a stick solves the problem of stability when walking over uneven ground or with an unsteady gait. Here virtualization serves a purpose. Similarly, the office is not replaced by telecommuting through any natural evolution, but because the company wants, say, to cut expenses or increase productivity – that is, because the company wants to maximize its profits, which is a social purpose (COVID-19, of course, imposed a different social purpose). Virtualizations also occur for less concrete goals, such as knowledge, self-expression, play, and pleasure.

One might ask if an envisioned virtuality is merely an act of imagination. Certainly imagination and creativity are involved in such envisioning, but an envisioned virtuality requires something more: a grounding relationship with materiality. The branch's materiality establishes *affordances* to a person who interacts with it – *pace* Lévy, strictly speaking the virtuality lies in their dynamic relationship rather than the branch itself. Likewise, although the Kanizsa triangle perception is a mental phenomenon, it nonetheless requires the affordances of the material image (in this case, the relationship between presence, absence, and the mind). One can imagine using a wheat stalk as a stick, but it does not offer the material affordances that could allow anyone to use it as a walking stick – no human, that is. Clustered and transmuted with other envisioned virtualities, it could become a walking stick for a tiny elf in a cartoon. (The instigating problem could be how to construct an interesting cartoon elf.) Here we touch on the relationship between virtuality and art, to be explored towards the end of this article.

How do these concepts play out in the human world? Imagine person X accepts an invitation from Y to meet for dinner. To get there, X has several considerations: What route is preferable, in terms of speed, simplicity, traffic, detours and such? By which means of transportation? What is the best departure time? Should there be allowances for the snow? These factors demarcate logical trajectories within a structured range of real possibilities, which is virtual because it delineates all of the viable paths from A to B. From that perspective the options' sole dissimilarity is that one of them – the route taken – becomes real and concrete, while the remainder vanish. The invitation itself involves virtuality: as someone once said, meeting for dinner is never just meeting for dinner. What history and potential future of X and Y's relationship prompted the invitation? Each individual factor above is a structure of real possibilities, a virtuality. Behind them are numerous large-scale virtualities (transportation structures, financial systems, weather dynamics, personal preferences) which may be actualized in various ways (e.g. traffic jams) that shape each other (because of the snow X will take the subway, but repair work on the local line is causing delays so the express line might be faster even though it requires more walking, except the sidewalks probably have not all been shoveled, so wear snow boots). Moreover, the intersection of multiple systems and structures creates turbulence and unpredictability, vastly expanding the range of possibilities itself. Things can go radically off course; the improbable, the surprising, the disastrous and the whimsical enter the scene. As Rutzou observes, critical realism does not typically think in these terms; to formulate a realism that recognizes reality as a 'complex combination of chaos and order', Deleuze is better equipped (Rutzou 2017, 415). That equipment partly derives from the mathematical models Deleuze draws upon, which became central to chaos theory (then nascent).

But what of the wide disparities among the variants of virtuality? The virtuality of an underlying, relationally structured, causal potential (the seed, the Kanizsa triangle, capitalism) differs from the inference of an original purpose (the office), which differs from an envisioned new purpose (the stick). And none of these appears related to the virtual as 'having the same or similar power or effect of something else' and 'a simulation', the dominant meanings despite Deleuze's assertions that actualization involves no similarity with the virtual. How do we coordinate these meanings, and does simulation fit into Deleuze's ontology at all?

To answer, we must consider that his is an ontology of becoming, consisting of the flow from virtualization to actualization to virtualization and so on, through time. Actualization as differenciation is a historical process (DeLanda 2002, 35–36) – but virtualization is as well. Virtuality as potentiality is located in the present, perhaps active, perhaps not; we can also trace an entity's nearest or most likely structural source, cause, function, or substratum, diachronically (or sometimes synchronically); and we can determine a potential future such as a new utility, or extrapolate a process's tendencies. Similarly for agents: our every act is born into preexisting conditions, a set of problems and solutions, a few of our own making; we seek the problem behind given solutions, solutions to given problems, and give ourselves problems to solve. Virtuality is inherent in all these problems. For agents the problem-solution, question-answer cycles are necessarily narrative, and the different senses of virtualization and actualization fluctuate around this historicity. But the present in the flow of potentiality, cause and projection is always future-past (here's what I'll do, here's what I did, any 'am doing' always already receding into the past); the sense of time itself is virtual, produced by memory. There is, however, another present, the present of potentiality in the moment of action or exertion, as though it were time itself. It is here that simulations – in Deleuze's terms, simulacra – can be produced. 'Simulacra are those systems in which different relates to different *by means of* difference itself' (Deleuze 1994, 299, his italics; see also 69, 277). Because 'the differential relation [is] the pure element of potentiality' (47), Deleuze is speaking here of virtualities. But the simulacrum's 'by means of' is not a mere repetition of virtuality: it is a *recursive* virtuality. (Deleuze does not use that term, but the idea is manifest at several locations.) The simulacrum lies pivotal in Deleuze's critique of Plato, for whom true reality consists of abstract, essential idealities (Deleuze uses the term 'models', somewhat confusingly for critical realists) which concrete things copy or represent. That ontology is threatened by the copy of a copy – the simulacrum, which is unlinked to ideality. We may say its recursiveness is the secret to that detachment. According to Deleuze, for Plato the treacherous simulacrum was the sophist; but Plato was barely less hostile to art, equally viewed as copies of copies, that is, as mimetic. To the extent that a simulacrum resembles something, Deleuze argues, the resemblance is an external effect which simulates something other than itself. We can see this, for example, in the mathematics of physical mechanics, where a displacement is used as a 'virtual velocity', and today the digital simulates the physical through 'virtual reality'; but as Deleuze's own use of mathematics as an ontological model (an analogue) demonstrates, a simulacrum need not create sensuous similarity at all. Yet despite his hedges, a simulacrum *must* have some sort of similarity to something or it could not function as a simulacrum, even if that similarity is born from the meanings of words or other signs. In the present argument, the simulacrum is pivotal as a 'potentialized potential': something that has the effect or structural 'equi=valency' of something else, including not just the obvious equi=valency borne by simulations, but also forms dense with meaning such as scientific models, symbols, metaphors, fiction, and the arts generally.[4] In sum, virtuality is a relational structure bearing a generative potential that can work in several ways, distinguishable as temporal moments in the ontology of becoming and generalizable as material problem-solution, question-answer cycles; and anything operating in such a manner has been cleaved with a lightning bolt of virtuality.

Is Deleuze's virtuality compatible with critical realism to any further extent? Bhaskar himself never discusses virtuality, but the possibility of congruence is suggested by the entry on virtuality in the *Dictionary of Critical Realism*:

> relations . . . are virtual or 'ideal' because, while requiring material things as their relata, they have no material substance and yet are sui generis real. . . . [I]n the sociosphere the virtual is best-conceived, not as contrasting with reality, but as constituting a sui generis real emergent stratum of reality with its own irreducible causal powers. (Hartwig 2007, 498–499; italics removed).

Given the lack of guidance from Bhaskar or previous critical realist analysis, the entry is a good first stab at the subject, clearly revealing overlaps with Deleuze, particularly in defining the virtual as both real and involving relationships grounded in material things – in a word, as structure, which perhaps is the point behind the scare quotes surrounding 'ideal'. Still, Deleuze's analysis improves it. First, elsewhere the *Dictionary*'s entry observes the tension between virtuality as having the actual effect of something else versus giving the appearance of something else. The array of Deleuzean virtualities – structurally causal, inferred, envisioned, and simulacral – resolves that tension by connecting them to moments in process ontology. Second, Deleuze provides a fuller concept of models, which is important for critical realism because, as Bhaskar argues (e.g. 1986, 68–70), models are central to scientific theorization, making them an essential part of epistemology.

Third, and most importantly, the *Dictionary* suggests that at least within the sociosphere the virtual constitutes an additional ontological stratum possessing its own causal powers – a supplement that a Deleuzean analysis avoids. The *Dictionary* does not state what set of ontological strata is at issue, or why it specifically concerns the sociosphere; possibilities include the pair social structure and agency; in Archer's version, structure, agency, and the cultural system; Bhaskar's concept of the social cube; or his four-planar social being. As there is no way to tell, I will turn to the most fundamental strata: Bhaskar's ontological domains of the real, the actual, and the empirical (Bhaskar 1978, 56–57). The real consists of powers, susceptibilities, generative mechanisms and structures – anything that is *causal* – that exist regardless of whether they are active. Notably, everything is either part of or itself a structure. Through their interactions real entities exercise or manifest their powers, producing the events that constitute the domain of the actual. As a result of such interactions, new generative mechanisms may emerge and others may evaporate. The third domain, the empirical, encompasses those events that humans experience; I will take it up later. The *Dictionary* contemplates adding a stratum to social ontology somewhere, consisting of relations, but that logic would also apply to the domains. Deleuze obviates such supplementation, because his interpretation of the virtual and the actual closely accords with critical realism's domains of the real and the actual. If in Deleuze's schema '[t]he reality of the virtual is structure' and virtuality as potentiality pertains to powers and generative mechanisms, then the virtual is tantamount to the domain of the real.[5] Physical powers establish virtualities in the sense of potent structured ranges of possibility, which involve tendencies (e.g. mass bends spacetime); and some relational structures produce non-physical and hence virtual powers (e.g. the Kanizsa triangle, social relationships). Although Deleuze eschews the concept of causality, actualization involves exactly the sort of interactions that produce the domain of the actual: the seed is a virtual tree, and its actualization depends on numerous contingencies and interactions – the seed has (or simply is) a generative mechanism that can in suitable circumstances grow into an actual tree. Critical realism bars conflating the domains of the real and the actual; likewise, Deleuze distinguishes the virtual and the actual, underscored by the difference and flow between actualization and virtualization. (Even when virtuality produces resemblances, such as simulations and models, the function of virtualization is to produce something similar but *not* the same in order to achieve a comparable effect, rejecting conflation.) Thus Deleuze's virtual/actual pair and critical realism's real/actual distinction are equivalent enough that the former can inform the latter or offer an alternative way to conceptualize it. In particular, Deleuze provides a way to conceptualize the powers of *relationships* rather than discrete entities, and the unfolding of virtuality into actualization. Since Deleuze's ontology grasps relationship, structure, and non-conflation, it can also support realist social ontology – social structures as real *and virtual* – without any ontological supplement.[6]

Conceptualizing social structures as virtual is not new: most notably, Anthony Giddens argues that '[t]o say that structure is a 'virtual order' of transformative relations means that . . . structure exists, as time–space presence, only in its instantiations in [social] practices and as memory traces orienting the conduct of knowledgeable human agents' (1984, 17). His concept of virtuality clearly differs from Deleuze's and it is unmistakably actualist. Archer contests Giddens's view that structure and agency are two sides of one thing. Her critique principally focuses on the conundrums created by his theory that social structures consist of rules and resources if these are supposed to be virtual (1995, 108–114). Her realist counter-theory is that structure and agency are two different things, each with its own set of properties. Regarding structures, '[w]hat differentiates a *structural* emergent property is its *primary dependence upon material resources*, both physical and human. In other words, the internal and necessary relations between its constituents are fundamentally material ones' (1995, 175; Archer's italics). Moreover, against Giddens's claim that rules exist solely as individuals' memory traces, Archer contends that they also exist in the form of laws, contracts, liturgies, and so on, independent of agents. (In addition, there is the extensive, relatively stable *social* memory embodied in enduring non-textual forms such as rituals, oaths, oral epics, symbolic objects, and family ties: memories in shared form. See e.g. Connerton (1989).) While this is right, a problem remains: what is the ontological status of relationships? Whatever their material dependencies or effects, relationships are not themselves physical. Relationships are virtual, hence both real and distinct from individuals; but because virtuality as potentiality must have material elements it does not pose the threat of idealism as Archer fears (1995, 110–111). Since social structures are systems of relationships (including relationships to material resources), one cannot escape their virtuality. Thus Giddens has a finger on a real aspect of social structures: they are indeed virtual, although not under his definition. Moreover, they are not instantiated but instead actualized through the interactions of social structures and agents.

Despite Deleuze's occasional use of virtuality and potentiality as synonyms, it is not clear whether they are actually synonymous or instead closely related. The latter seems preferable. Given the mathematical elements of Deleuze's philosophy and assertions such as '[t]he reality of the virtual is structure' (1994, 209) as well as examples such as the Kanizsa triangle, virtuality's emphasis is on relational composition. Potentiality is more commonly associated with causal powers, which Bhaskar embraces, but Deleuze keeps causality at arm's length or farther (e.g. 1994, 23; see also Bryant 2011, 102). Certainly, Bhaskar's generative mechanisms can be at least partly relational, but his focus remains on their powers. However, given the similar roles the two philosophers assign to process and contingency, virtuality and potentiality may constitute different aspects of entities and tendencies (i.e. the real), not different things or states.

As Rutzou observes, 'the realism of [a] hybrid Bhaskar-Deleuze rests in saving the primacy of difference, heterogeneity, and process as basic properties of any open system and proceeds from this base' (2017, 414). While I do not propose a hybrid so much as selective adoption and adaptation, this position seems right. However, Rutzou overstates a contrast between Deleuze as process-oriented and Bhaskar as object-oriented (2017, 409). (Ignore the confusing fact that some 'object-oriented' ontologists draw much from Deleuze and little from Bhaskar.) The overstatement largely derives from his decision to discount Bhaskar's *Dialectic* (1993) (see Rutzou 2017, 416n20). *Dialectic* gives an important place to several processual concepts, and repeatedly refers to being as becoming – indeed, the first of its climatic two sentences declares, 'to exist is to be able to become' (1993, 385; see also 77, 189, 198). Conversely, virtualities must be connected to objects, and actualizations always produce objects, though not necessarily physical ones. The contrast between Deleuze and Bhaskar (at least the Bhaskar of *Dialectic*) is not so stark.

Adapting the concept of virtuality for critical realism does require two clarifications or adjustments. First, as Levi Bryant observes, Deleuze is torn between considering the virtual a single continuum out of which all else arises (monism), versus a part of objects' substance individually (a pluralistic view); only the latter is viable from a realist perspective (Bryant 2011, 94–104). Second, in an argument leaning towards the pluralistic view, Deleuze writes, '[e]very object is double without it being the case that the two halves resemble one another, one being a virtual image and the other an actual image' (1994, 209). Taken out of context such statements might suggest that the virtual exists only when it has been actualized, which would be actualist. If the virtual consists of powers and capacities, it must exist with or without actualization, transfactually; actualization occurs through interactions between entities.

So far I have discussed how Deleuze's concepts of the virtual and the actual are tantamount to the domains of the real and the actual. *Difference and Repetition* does not propose anything akin to the empirical domain, but virtuality is relevant to the latter. Elsewhere I have argued that the empirical domain should be reconceptualized as the semiosic domain, defining it not through the concept of human experiences, but instead through the more precise and encompassing concept of signs – with the important understanding that not all signs are linguistic or other socially constructed symbols (Nellhaus 1998; 2024).[7] For instance, smoke is a sign of a fire, and a growling stomach might be a sign of hunger. A sign is a relationship between a signifier, signified, and referent (Bhaskar 1993, 223).[8] As I elaborate, '[s]igns require one or more physical support structures in order to exist, such as sound, marks and neurons, but their signifying power derives from their relational structure and can't be reduced to synaptic firings' (2024, 10). That relationship is virtual, not physical. In other words, ontologically, a sign exists and exercises its power virtually, as do entire sign systems such as language. The mind itself is virtual, just as meaning is – in a sense, more so, insofar as we describe the mind doing (or not doing) things *as if* it were a physical entity, through metaphors of wheels turning, being rusty on a subject, being like a steel trap, and so on: the *as if* of simulacral virtuality. At the same time the brain and embodiment as a whole deeply shape experience and make possible not only thoughts about the directly present, but also things not present, such as the past, the future, missing objects, imaginary objects, and counterfactuals. Thus human experience is built on a double structure consisting of the mind–brain system and signs, both of which depend on virtuality.

To my knowledge, Bhaskar's most extensive discussion of ideas is his article 'On the Ontological Status of Ideas' (1997), where he considers whether ideas are real and whether different types of ideas have different kinds of reality. The article unfortunately seems hastily written, as it sometimes drifts from its argument and makes a few uncharacteristic claims. That said, one of its key points is that ideas are real because they are causally efficacious. Moreover, ideas are 'emergent parts of the natural world system' (1997, 143); they are both irreducible to the conditions of their production and able to act back upon them. As we have seen, ideas differ from most entities emergent from the natural world because their powers reside strictly in their virtuality.

But for present purposes Bhaskar's most cogent argument is that 'tools, machines etc., cannot be conceived simply as material objects, but are also intrinsically the objectification of (socially produced and transformed) ideas' (1997, 144). The point is vital. The architecture of houses, for instance, altered under changed ideas of the family. Moreover, there is more to design than functionality: it influences our relationship with tools, machines, buildings, and other objects. People often notice the beauty of certain antique equipment because they incorporate functionally gratuitous design elements; the austere aesthetics of an assembly line accords with capitalism's intense squeeze on labour power. Cars are designed and purchased not only as transportation but also as symbols of power, status, and/or sex appeal; one reason behind the iPhone's popularity is its elegance. Many ideas consist of or are surrounded by clouds of images, which can be essential (the word 'sex' is not sexy, but a high-heeled shoe might be), and they are actualized most intensely in art. This capacity for polysemy is inherent in signs' virtuality, especially since signs can have multiple connections to materiality and vice versa.

But a translation back from images to words is not always possible or desirable. What does Picasso's *Guernica* 'say'? One can discuss the ideas that went into it fairly readily, but what it 'says' propositionally is trite in the sense that 'War is hell' tells us nothing new. What the painting *does* is something else entirely – yanking us from one scene to another, the movement from right to wrong converted into movement from right to left, twisting, distorting and ripping the world apart, simultaneously flattening it into drab greys. What it 'says' is performative: its *effects*. Virtuality is not static, it is a potential or power even when dormant, lying in wait, that can be actualized (exercised) materially. Recall my example, 'Chocolate is a virtual drug'. Requiring and affecting materiality, earthly dirt often clings to virtuality's roots. Nowhere is that truer than in simulacra, particularly those that produce simulations rather than models – and for Deleuze, 'art is simulation' (1994, 293). Although ideas *qua* ideas are all virtual, there are nevertheless ontological differences between simulacral ideas, with their recursive structure, and propositions, which are non-simulacral ideas. The two differ in their internal relational structure and their external relationships to materiality, other ideas, and history; that is, they are distinctive partial totalities as Bhaskar defines them (1993, 126). These ontological differences entail different *effects*, such as the impact of *Guernica* versus an assertion distilling it.

But partial totalities do not all weigh their internal and external relationships equally – they are not necessarily balanced halves, and as recursive structures, simulacra both intensify that inequality and instill in each part a different degree of openness or closure. Art, with its fictionality (see Walton 1990), and models, with their varying referential adequacy, are both simulacra, but they are simulacra in differing ways. For some simulacra, the external relations dominate: these are models, which seek logical internal coherence and may possess a certain beauty (as mathematicians often aver, and histories can be beautifully written), but insofar as they model things in the world, their truth or adequacy is primary and evaluated by their external relations. This makes the models comparatively abstract or 'disembodied' because their internal relations have weaker power.

For other simulacra, internal relations dominate: this is the fictionality of art, which is its primary virtuality, constituting artworks as their own worlds obeying their own rules. Novels predominantly depend on their own text, *Guernica* on its imagery. Music too creates a space of its own, into which we enter. Simulacra may have little in common with propositional truth in the sense of verisimilarity, as exemplified by caricatures, parodies, and non-realistic aesthetics generally.[9] As Lévy writes, art 'virtualizes virtualization because it attempts to supply both an escape from the here and now and its sensual exaltation' (1998, 99). If virtuality is always anchored in materiality, that can go double for art. (With text, that means its imagery and sometimes sounds.) One can assess and contextualize artworks in relation to their history and present, yet during engagement, to a greater or lesser extent one 'detaches' from the weakened external world to immerse oneself in the fictional one, to be drawn into it – notice the metaphors of internality – fostering a concrete embodied connection in which one's individual (and culturally shaped) capacities, experience and assumptions play critical roles. The extent of artworks' inward closure can vary: for instance, instrumental music and abstract painting have stronger intrinsic inward closure than theatre because they depend heavily on internal relations, whereas theatre's closure is highly leaky due to the audience-performer dynamic, allowing (say) direct address to the audience. Nevertheless, the internal relations are stronger than the external ones. Rather than serve the actual world as models do, art makes virtual worlds, under the sense of virtuality as potentiality. The locus of 'real possibility' shifts to inside the artwork itself. We step into such a world in *Guernica*, and likewise in *Star Trek*. To theorize art as simulacra, and therefore inherently fictional, critical realism must reject the concept of art as mimesis (subordinate to an original), derived from Plato by way of Aristotle; nor can formalism, expressivism or phenomenology explicate this side of art. Instead, realists must understand art's fictionality as world-construction.[10] In narrative arts such as novels, plays and movies, and sometimes in other arts, these virtual worlds can be conceptualized as possible worlds (Doležel 1998; Ryan 2015, 69–75). And if we can shed the connotations of falsity and deception from the concepts of the virtual and the simulacrum, then we can do the same for the term describing a simulacrum's action and its present-tenseness: it is *performative*, in the sense of production and efficacy – in tune with its position within an ontology of becoming. Most of *Guernica*'s meaning consists of what it performs upon us.

Simulacra of all types are performative by definition. But as we see, artistic and scientific simulacra have opposite frames of reference, with art's inherent fictionality requiring a 'cognitive relocation' to its own reality. Ryan argues that if a biography of Shakespeare asserts that he was born in Stratford, England, external evidence buttresses that contention; but King Lear had three daughters, as evidenced internally by the play *King Lear* upon Shakespeare's say-so and his alone. She describes the difference as an 'indexical' definition of referential truth, pointing reference to outside or inside a text. However, this concerns the referential logic, not the experiential contact – histories and biographies can transfix the imagination as fully as fiction, without referential reorientation (Ryan 2015, 71–75, 98).[11] The ontological distinction between simulacra thus has epistemological consequences, as reference points the simulacrum outward or inward. Referential reorientation is reflected in simulacra's contrasting navigation of the question/answer dialectic: models ask questions that seek an answer for validation; artworks typically ask questions that may linger without an answer – performative effects are foremost. To be clear, cognitive relocation can be quite slim and brief, as we will see. Also, not any virtual totality is a simulacrum: within formal logic and pure mathematics, for example, formulas are intentionally barren of referential content; they are governed by definitions of validity, but say nothing true or false of any world.[12] Nearly all philosophy (including critical realism) is not formal in this sense, nor does much of it transport one to some other space.

But some philosophical texts do transport, and to return us to the beginning, they include the writings by the original philosopher of flux, Heraclitus, what few scraps of it are extant. Norrie devotes considerable attention to how Bhaskar and Deleuze understood Heraclitus; however, I am interested less in his writing's propositional content than its performative effects. His writings stand out for their literary, even poetic quality, potent with ambiguities, a style that influenced subsequent generations of classical philosophers; after all, Aeschylus and Pindar were his contemporaries, during an era when poetry and philosophy often walked together (Kahn 1979, 3–8). This density appears in his best-known fragment, '*potamoisi toisin autoisin embainousin hetera kai hetera hudata epirrei*',[13] which in Charles H. Kahn's nearly literal translation reads, 'As they step into the same rivers, other and still other waters flow upon them' (1979, 53). For a flickering moment, we must imagine flowing rivers, with people stepping into them. But *are* rivers what do and do not stay the same? As Kahn points out, Heraclitus' ambiguous syntax permits another reading; ultimately, both rivers and people remain the same yet constantly change. The words themselves create auditory rivers, with the first phrase's burbling sibilants and diphthongs, and the second phrase's repetition of 'other' (*hetera*) (Kahn 1979, 167). The polysemy and literary devices pull the reader more closely into the imagery – and literary style becomes part of Heraclitus' philosophical argument. As Daniel Graham (2023) observes, 'his *logoi* [i.e. words, thoughts] are designed to be experienced, not just understood, and only those who experience them in their richness will grasp his message'. Heraclitus' ambiguous phrasing, for instance, produces a soft shuttling between meanings, not distant from another theme in his philosophy, the cyclic cosmos. As if the river-line were also an annulus, sans annulment, the text's own simulacrum: a structured flux. The meaning of what he says arrives partly through what his writing performs. So too might an article written as an unsectioned stream.

Deleuzean virtuality as potentiality, as adopted and adapted here with all its several meanings, offers critical realism new perspectives on its own theories, and concepts that can expand its purview – in particular, by providing concepts through which we can understand powers that are purely relational. The virtual/actual and virtualization/actualization couplets present another way to think about powers and their interactions, that is, the real and the actual as critical realism has interpreted them. Virtuality brings a better understanding of signs (the fundamental components of the semiosic/empirical domain) as not just conjoining signifiers, signifieds, and referents, but also as the relationships between and among them, which, because the relationships are virtual, are imbued with causal power. And virtuality provides critical realism with concepts for understanding at least part of the ontology of models and art.

Notes

1. There are realist and non-realist interpretations of Deleuze. DeLanda's account is especially compelling and obviously that view of Deleuze makes him more available for critical realism.
2. Deleuze often uses biological examples of differenciation and actualization, yet bafflingly, Norrie accuses him of lacking 'a sense of how change is actualised' (2010, 205). See also DeLanda (2002).
3. Strictly speaking, virtualization extends to the natural world, as evolution involves problem-solution cycles that expand mobility, environmental flexibility, etc. (Lévy 1998, 31–32). The light 'problem' that is 'solved' by creating eyes is an example.
4. Actually, Deleuze asserts that *everything* is a simulacrum, not as illusions à la Baudrillard, but as entities that are not 'copies' or 'imitations' of an ideal. I think this defuses the concept.
5. Similarly, see also Bryant (2011, 95, 113) from a Deleuzean perspective, and Rutzou (2015, 189, 232–233) from within critical realism, although he leans toward viewing Deleuze's account as the stronger.
6. Deleuze's foundational concept of difference also has kinship with absence, to which Bhaskar (2008) accords ontological primacy; but analysis of this relationship and the role of what Deleuze calls '(non)-being' or '?-being' lies beyond this article's scope. Norrie offers an important critique (2010, 195–197, 204–205), although by ignoring virtuality he forecloses a more positive assessment.
7. The redefinition also forecloses anthropocentrism (see Nellhaus 2024).
8. The concepts signifier, signified, and referent should be replaced by Pierce's more sophisticated counterparts (Nellhaus 2024), but the issues need not detain us.
9. Aesthetic realism should not be conflated with philosophical realism – sometimes it is even irrealist.
10. Also called world-building, world-creation, etc.; see Doležel (1998, x, 1–28). Walton (1990) treats art's fictionality as mimetic, an idealist account. Wilson (2020) attempts (not altogether

successfully) a critical realist theory of aesthetic experience rather than art *per se*, and consequently does not address mimesis.
11. The indexical view of referential truth derives from David Lewis.
12. It is tempting to say such totalities are completely closed, but Gödel's incompleteness theorems demonstrate that no sufficiently complex formal system in logic can prove its own consistency, and some true statements cannot be proven within that system; proof, if any, must come from a more encompassing system, which has the same incompleteness. Totalities tend to leak.
13. Ποταμοῖσι τοῖσιν αὐτοῖσιν ἐμβαίνουσιν ἕτερα καὶ ἕτερα ὕδατα ἐπιρρεῖ.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Tobin Nellhaus http://orcid.org/0000-0003-3622-0374

References

Archer, Margaret S. 1995. *Realist Social Theory: The Morphogenetic Approach*. Cambridge, UK: Cambridge University Press.
Archer, Margaret S. 2003. *Structure, Agency and the Internal Conversation*. Cambridge, UK: Cambridge University Press.
Archer, Margaret S. 2013. "Collective Reflexivity: A Relational Case for It." In *Conceptualizing Relational Sociology: Ontological and Theoretical Issues*, edited by Christopher Powell, and François Dépelteau, 145–161. New York: Palgrave Macmillan.
Bhaskar, Roy. 1978. *A Realist Theory of Science*. Hassocks, Sussex; Atlantic Highlands, NJ: Harvester Press; Humanities Press.
Bhaskar, Roy. 1986. *Scientific Realism and Human Emancipation*. London: Verso.
Bhaskar, Roy. 1993. *Dialectic: The Pulse of Freedom*. London: Verso.
Bhaskar, Roy. 1997. "On the Ontological Status of Ideas." *Journal for the Theory of Social Behaviour* 27 (2–3): 139–147. https://doi.org/10.1111/1468-5914.00031.
Bryant, Levi R. 2011. *The Democracy of Objects*. Ann Arbor, MI: Open Humanities Press.
Connerton, Paul. 1989. *How Societies Remember*. Cambridge, UK: Cambridge University Press.
Danermark, Berth, Mats Ekström, and Jan Ch. Karlsson. 2019. *Explaining Society: Critical Realism in the Social Sciences*. 2nd ed. Abingdon, Oxon, UK; New York: Routledge.
DeLanda, Manuel. 2002. *Intensive Science and Virtual Philosophy*. London: Continuum.
DeLanda, Manuel. 2011. "Emergence, Causality and Realism." In *The Speculative Turn: Continental Materialism and Realism*, edited by Levi Bryant, Nick Srnicek, and Graham Harman, 381–392. Melbourne: Re.Press. https://re-press.org/books/the-speculative-turn-continental-materialism-and-realism/.
DeLanda, Manuel, and Graham Harman. 2017. *The Rise of Realism*. Malden, MA: Polity.
Deleuze, Gilles. 1994. *Difference and Repetition*. Translated by Paul Patton. New York: Columbia University Press.
Doležel, Lubomír. 1998. *Heterocosmica: Fiction and Possible Worlds*. Baltimore: Johns Hopkins University Press.
Giddens, Anthony. 1984. *The Constitution of Society: Outline of the Theory of Structuration*. Cambridge, UK: Polity Press.
Graham, Daniel W. 2023. "Heraclitus." In *The Stanford Encyclopedia of Philosophy (Winter 2023 Edition)*, edited by Edward N. Zalta, and Uri Nodelman. Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/win2023/entries/heraclitus/.

Hartwig, Mervyn, ed. 2007. *Dictionary of Critical Realism*. London: Routledge.
Kahn, Charles H. 1979. *The Art and Thought of Heraclitus: An Edition of the Fragments with Translation and Commentary*. Cambridge, UK: Cambridge University Press.
Lévy, Pierre. 1998. *Becoming Virtual: Reality in the Digital Age [Qu'est-Ce Que Le Virtuel?]*. Translated by Robert Bononno. New York: Plenum.
Massumi, Brian. 2014. "Envisioning the Virtual." In *The Oxford Handbook of Virtuality*, edited by Mark Grimshaw, 55–70. Oxford: Oxford University Press.
Nellhaus, Tobin. 1998. "Signs, Social Ontology, and Critical Realism." *Journal for the Theory of Social Behaviour* 28 (1): 1–24. https://doi.org/10.1111/1468-5914.00060.
Nellhaus, Tobin. 2024. "Peirce's Universal Categories and Critical Realist Ontology." *Journal for the Theory of Social Behaviour* 55 (1): e12435. https://doi.org/10.1111/jtsb.12435.
Norrie, Alan. 2010. *Dialectic and Difference: Dialectical Critical Realism and the Grounds of Justice*. London, New York: Routledge.
Rutzou, Timothy. 2015. "In Pursuit of the Real: Postmodernism and Critical Realism". PhD, London: University College London. https://discovery.ucl.ac.uk/id/eprint/10021917/.
Rutzou, Timothy. 2017. "Finding Bhaskar in All the Wrong Places? Causation, Process, and Structure in Bhaskar and Deleuze." *Journal for the Theory of Social Behaviour* 47 (4): 402–417. https://doi.org/10.1111/jtsb.12138.
Rutzou, Timothy, and Dave Elder-Vass. 2019. "On Assemblages and Things: Fluidity, Stability, Causation Stories, and Formation Stories." *Sociological Theory* 37 (4): 401–424. https://doi.org/10.1177/0735275119888250.
Ryan, Marie-Laure. 2015. *Narrative as Virtual Reality 2: Revisiting Immersion and Interactivity in Literature and Electronic Media*. Baltimore: Johns Hopkins University Press.
Walton, Kendall L. 1990. *Mimesis as Make-Believe: On the Foundations of the Representational Arts*. Cambridge, MA: Harvard University Press.
Wilson, Nick. 2020. *The Space That Separates: A Realist Theory of Art*. Abingdon, Oxon, UK: Routledge.